Домашняя работа №1

№1

$\vec{X} = (X_1, \dots, X_n)$, $X_i \sim Bin(m, p)$, m - изв. p - неизв.

а) $\vec{X}$ - статистика б) $m^2 + p$ - нет, p - не знаем

в) $X_{(n)}$ - статистика г) $X_1 + X_2 + 2$ ✓

д) $\sqrt{m} \sum_{k=1}^{n} X_k$ - ✓ е) $\prod_{k=1}^{n} X_k^p$ - нет, ж) 2025 ✓

№2

$\vec{X} = (X_1 \dots X_n)$, $X_i \sim F$ с плотн. $f(x)$

а) $X_{(n)}$; $F_{X_{(n)}}(x) = P(X_{(n)} < x) = P(X_{(1)} < x, X_{(2)} < x \dots$
$\dots, X_{(n)} < x) = P(X_1 < x, X_2 < x, \dots, X_n < x) = \prod_{i=1}^{n} P(X_i < x) =$

$= (P(X_n < x))^n = F_{X_n}^n(x)$

$f_{X_{(n)}}(x) = F'_{X_{(n)}}(x) = (F_{X_n}^n(x))' = n F_{X_n}^{n-1}(x) \cdot f(x)$

б) $X_{(1)}$

$F_{X_{(1)}}(x) = P(X_{(1)} < x) = 1 - P(X_{(1)} > x) = 1 -$
$- P(X_{(1)} > x, X_{(2)} > x, \dots, X_{(n)} > x) = 1 - P(X_1 > x, \dots, X_n > x) =$
$= 1 - P(X_1 > x)^n = 1 - (1 - P(X_1 < x))^n = 1 - (1 -$
$- F_{X_1}(x))^n$; $f_{X_{(1)}}(x) = F'_{X_{(1)}}(x) = -n(1 - F_{X_1}(x))^{n-1} \cdot$
$\cdot (-f(x)) = n f(x) (1 - F_{X_1}(x))^{n-1}$

в) $X_{(k)}$, $1 < k < n$

$F_{X_{(k)}}(x) = P(X_{(k)} < x) = P(X_{(1)} < x, \dots, X_{(k)} < x)$

$\exists$ $k+i$ величин $< x$, $n-i-k$ величин - больше - событие A

~~$A(X_{(1)} < x, \dots, X_{(k+i)} < x, X_{(k+i+1)} > x, \dots X_{(n)} > x) =$~~

~~$P(X_{(1)} < x, \dots, X_{(k)}) = 1 - P(X_{(1)} > x, \dots, X_{(k)} > x) =$~~
~~$= 1 - P(X_{(1)} > x, \dots, X_{(n)} > x) = 1 - \prod_{i=1}^{n} P(X_{(i)} > x)$~~

~~$P(X_{(1)} < x, \dots, X_{(k)} < x) = P(X_{(1)} < x, \dots, X_{(k+i)} < x) = 1 -$~~
~~$- P(X_{(1)} > x, \dots$~~

$P(A_i) = P(X_{(1)} < x, \dots, X_{(k+i)} < x, X_{(k+i+1)} > x, \dots, X_{(n)} > x) =$

$= C_n^{k+i} \prod_{j=1}^{k+i} P(X_1 < x) \cdot \prod_{j=k+i+1}^{n} P(X_1 \geq x)$

Событие $(X_{(1)} < x \dots X_{(k)} < x) = \bigcup_{i=0}^{n-k} A_i$

$P(X_{(1)} < x, \dots, X_{(k)} < x) = \sum_{i=1}^{n-k} C_n^{k+i} P(X_1 < x) P(X_1 \geq x) =$

$= \sum_{i=1}^{n-k} C_n^{k+i} (P(X_1 < x))^{k+i} (P(X_1 \geq x))^{n-k-i} = \sum_{i=0}^{n-k} C_n^{k+i} F_{X_1}^{k+i}(x) (1 - F_{X_1}(x))^{n-k-i}$

$F_{X_{(k)}}(x) = \sum_{i=0}^{n-k} C_n^{k+i} F^{k+i}(x) (1 - F(x))^{n-k-i} = \sum_{i=k}^{n} C_n^i F^i(x) (1 - F(x))^{n-i} =$

$\cancel{f_{X_{(k)}}(x) = \sum_{i=1}^{n-k} C_n^{k+i} (k+i) F^{k+i-1}(x) \cdot (n-k-i)(1 - F(x))}$

$\cancel{= \sum_{i=0}^{n} C_n^i F^i(x) (1 - F(x))^{n-i} - \sum_{i=0}^{k} C_n^i F^i(x) (1 - F(x))^{n-i}}$

$\cancel{= (F(x) + 1 - F(x))}$ $f_{X_{(k)}} = \sum_{i=k}^{n} C_n^i \, i(n-i) F^{i-1}(x) (1 - F(x))^{n-i-1} \cdot$

$(-f^2(x))$.

№4

$\vec{X} = (X_1, \dots, X_n)$, $X_i \sim Geom(p)$. $P(X = m) = (1-p)^m p$

$\overline{X}$ - достат. статист. для p?

$L(X_1, \dots, X_n, p) = P(X_1 = x_1) \cdot P(X_2 = x_2) \cdot \dots \cdot P(X_n = x_n) \xrightarrow{max} =$

$= (1-p)^{x_1} p \cdot (1-p)^{x_2} p \cdot \dots \cdot (1-p)^{x_n} p = p^n (1-p)^{\sum x_i} \longrightarrow max$

$L'(p) = np^{n-1} (1-p)^{\sum x_i} - p^n \sum x_i (1-p)^{\sum x_i - 1} = 0$

$\cancel{np^{n-1} (1-p)^{\sum x_i} = p^n \sum x_i (1-p)^{\sum x_i - 1}}$

$\cancel{n(1-p) = p \sum x_i, \; p \sum x_i + np = n, \; p = \frac{n}{n + \sum x_i}}$

$\frac{1}{p} = \frac{n + \sum x_i}{n} = 1 + \overline{X}$, $p = \frac{1}{1 + \overline{X}}$

L_{max} = Согласно критерию факторизации, если $L = g(\overline{X}, p) \cdot h(\vec{X})$, то статистика достаточно

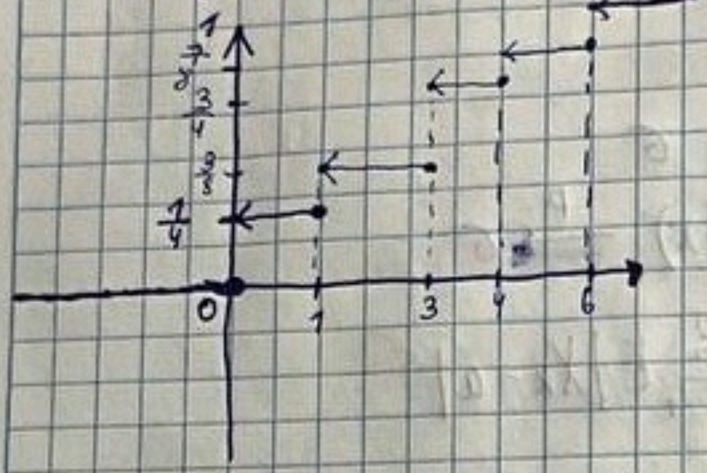
Подобрать такую функцию нельзя, т.к. $\sum x_i = n\overline{X}$ — зависимость еще и от $n \Rightarrow$

$\Rightarrow$ статистика $\overline{X}$ не является достаточной.

№3

$(3, 0, 4, 3, 6, 0, 3, 1)$ – значения выборки

$$F_n^*(y) = \frac{1}{n}\sum_{i=1}^{n} I(X_i < y)$$



$F_8^*(1) = \frac{2}{8} = \frac{1}{4}$ (2 нуля из 8 чисел)

$F_8^*(3) = \frac{3}{8}$ (2 нуля, 1 единица)

$F_8^*(5) = \frac{7}{8}$ (не хватает только 6)

№5

$X_i \sim U[0, \theta]$, $\vec{X} = (X_1, \dots, X_n)$

$X_{(n)}$ дает. ассимп. несмещ оценку θ, т.е

$$\lim_{n\to\infty} EX_{(n)} = \theta$$

Найдем функцию распределения $X_{(n)}$

$$F_{X_{(n)}}(x) = P(X_{(n)} < x) = P(X_{(1)} < x, \dots, X_{(n)} < x) = P(X_1 < x, \dots, X_n < x) =$$

$$= \prod_{i=1}^{n} P(X_1 < x) = F_x^n(x)$$

Плотность распределения: $f_{X_{(n)}}(x) = (F_x^n(x))' = n F_x^{n-1}(x) \cdot f_x(x) =$

$\cancel{= n \cdot \left(\frac{x}{\theta}\right)^{n-1}}$

$$F_x^n(x) = \begin{cases} 0 & x < 0 \\ \left(\frac{x}{\theta}\right)^n, & 0 \le x \le \theta \\ 1, & x > \theta \end{cases} \qquad f_{X_{(n)}}(x) = \begin{cases} 0, & x < 0 \\ n \cdot \left(\frac{x}{\theta}\right)^{n-1} \cdot \frac{1}{\theta}, & x \in [0, \theta] \\ 0, & x > \theta \end{cases}$$

$\cancel{EX_{(n)} = \int_0^\theta \frac{n}{\theta^n} x^{n-1} \cdot x\,dx = \frac{n}{\theta^n} \frac{x^n}{n} \Big|_0^\theta =}$

$$EX_{(n)} = \int_0^\theta f_{X_{(n)}}(x) \cdot x\,dx = \int_0^\theta \frac{n}{\theta^n} x^{n-1} \cdot x\,dx = \frac{n}{\theta^n} \cdot \frac{x^{n+1}}{n+1}\Big|_0^\theta =$$

$$= \frac{n}{n+1}\theta$$

$\lim\limits_{n\to\infty} EX_{(n)} = \lim\limits_{n\to\infty} \frac{n}{n+1}\theta = \theta = \theta \Rightarrow X_{(n)}$ – ассимп. несмещ. оценка θ ч.т.д.

$$EX^2 = DX + (EX)^2 = \frac{1}{2}(\lambda_1+\lambda_2) + \frac{1}{4}(\lambda_1+\lambda_2)^2$$

$$E\hat{X} = \bar{X} \qquad \begin{cases} \bar{X} = 0{,}5\lambda_1 + 0{,}5\lambda_2 \\ \overline{X^2} = 0{,}25\lambda_1^2 + 0{,}25\lambda_2^2 + 0{,}5\lambda_1\lambda_2 - 0{,}5\lambda_1 + 0{,}5\lambda_2 \end{cases}$$

$$EX^2 = \overline{X^2}$$

$$\lambda_2 = 2\bar{X} - \lambda_1$$

$$\bar{X}^2 = 0{,}25\lambda_1^2 + 0{,}25(2\bar{X}-\lambda_1)^2 + 0{,}5\lambda_1(2\bar{X}-\lambda_1) + 0{,}5\lambda_1 + 0{,}5(2\bar{X}-\lambda_1)$$

$$EX^2 = \sum_{k=1}^{\infty} k^2 P(X=k) = \frac{1}{2}\sum_{k=1}^{\infty} k^2 \cdot \left(\frac{\lambda_1^k}{k!}e^{-\lambda_1} + \frac{\lambda_2^k}{k!}e^{-\lambda_2}\right) = \frac{1}{2}\sum k^2 \frac{\lambda_1^k}{k!}e^{-\lambda_1} +$$

$$+ \frac{1}{2}\sum k^2 \frac{\lambda_2^k}{k!}e^{-\lambda_2} = \frac{1}{2}(\lambda_1 + \lambda_1^2) + \frac{1}{2}(\lambda_2 + \lambda_2^2)$$

$$EX = \bar{X}, \quad EX^2 = \overline{X^2}$$

$$\begin{cases} \bar{X} = \frac{1}{2}\lambda_1 + \frac{1}{2}\lambda_2 \\ \overline{X^2} = \frac{1}{2}\lambda_1 + \frac{1}{2}\lambda_2 + \frac{1}{2}\lambda_1^2 + \frac{1}{2}\lambda_2^2 \end{cases} \qquad \lambda_2 = 2\bar{X} - \lambda_1$$

$$\overline{X^2} = \bar{X} + \frac{1}{2}\lambda_1^2 + (2\bar{X}-\lambda_1)^2 \cdot \frac{1}{2} = \bar{X} + \lambda_1^2 + 2\bar{X}^2 - 2\bar{X}\lambda_1$$

$$\lambda_1^2 - 2\bar{X}\lambda_1 + 2\bar{X}^2 - \overline{X^2} + \bar{X} = 0$$

$$D = 4\bar{X}^2 - 8\bar{X}^2 + 4\overline{X^2} - 4\bar{X} = 4\overline{X^2} - 4\bar{X}^2 - 4\bar{X}$$

$$\lambda_1 = \bar{X} \mp \sqrt{\overline{X^2} - \bar{X}^2 - \bar{X}} \qquad \lambda_1 < \lambda_2$$

$$\lambda_2 = 2\bar{X} - \lambda_1 = 2\bar{X} - \bar{X} \pm \sqrt{\overline{X^2} - \bar{X}^2 - \bar{X}} = \bar{X} \pm \sqrt{\overline{X^2} - \bar{X}^2 - \bar{X}}$$

Отсюда: б) $\bar{X} = \frac{28\cdot 0 + 1\cdot 47 + 2\cdot 81 + 3\cdot 67 + 4\cdot 53 + 5\cdot 24 + 6\cdot 13 + 7\cdot 8 + 8\cdot 3 + 9\cdot 2 + 10\cdot 1}{327}$

$$= \frac{928}{327} = \frac{29\cdot 32}{3\cdot 109}$$

$$\overline{X^2} = \frac{28\cdot 0^2 + 1^2\cdot 47 + 2^2\cdot 81 + 3^2\cdot 67 + 4^2\cdot 53 + 5^2\cdot 24 + 6^2\cdot 13 + 7^2\cdot 8 + 8^2\cdot 3 + 9^2\cdot 2 + 10^2\cdot 1}{327} = \frac{3736}{327} =$$

$$\lambda_1 = \frac{928}{327} \pm \sqrt{\frac{3736}{327} - \frac{928}{327} - \frac{928^2}{327^2}} \approx 2{,}1$$

$$\lambda_2 \approx 3{,}57$$

Ответ: $\lambda_1 \approx 2{,}1$, $\lambda_2 \approx 3{,}57$

N8

1) Geom(p)

$$P(X_i = k) = p\cdot(1-p)^{k}$$

$$L(p) = P(X_1 = k_1)\cdot\ldots\cdot P(X_n = k_n) = p^n(1-p)^{\sum k_i}$$

$$\ln L(p) = n\cdot\ln p + \sum_{i=1}^{n} k_i \ln(1-p)$$

$$\frac{\partial \ln L(p)}{\partial p} = \frac{n}{p} - \frac{\sum k_i}{1-p} = 0, \quad (1-p)n = \sum k_i \cdot p, \quad p\left(\sum k_i + n\right) = 1$$

$$p^* = \frac{1}{\sum k_i + n}$$

$$\frac{\partial^2 \ln L(p)}{\partial^2 p} = -\frac{n}{p^2} - \frac{\sum x_i}{(1-p)^2} = \left| p^* = \frac{1}{\sum x_i + n} \right| = -\frac{n}{(\sum x_i + n)^2} - \frac{\sum x_i}{(1-\sum x_i \cdot n)^2} < 0 \Rightarrow$$

$\Rightarrow p^* = \arg\max$ при $p^* = \frac{1}{\sum x_i + n}$

2) $E(\alpha)$

$$f(x) = \alpha e^{-\alpha x} I\{x \geq 0\}$$

$$L(\alpha) = \prod_{i=1}^{n} f(x_i) = \alpha^n e^{-\alpha \sum x_i}$$

$$\ln L(\alpha) = n \ln \alpha - \alpha \sum x_i$$

$$\frac{\partial \ln L(\alpha)}{\partial \alpha} = \frac{n}{\alpha} - \sum x_i = 0, \quad \alpha^* = \frac{n}{\sum x_i} = \frac{1}{\bar{x}}$$

$$\frac{\partial^2 \ln L(\alpha)}{\partial^2 \alpha} = -\frac{n}{\alpha^2} < 0 \Rightarrow \alpha^* = \arg\max$$

Ответ: $p^* = \frac{1}{\sum x_i + n}$, $\alpha^* = \frac{1}{\bar{x}}$

№9

$$f(x) = \frac{\beta \theta^\beta}{x^{\beta+1}}, \quad x \geq \theta$$

$$L(\beta, \theta) = \prod_{i=1}^{n} f(x_i) = \frac{\beta^n \theta^{n\beta}}{\prod_{i=1}^{n} x_i^{\beta+1}}$$

$$\ln L(\beta, \theta) = n \ln(\beta \theta^\beta) - (\beta+1) \ln\left(\prod_{i=1}^{n} x_i\right) =$$

$$= n \ln \beta + n\beta \ln \theta - (\beta+1) \sum_{i=1}^{n} \ln x_i$$

а) θ известно

$$\frac{\partial \ln L(\beta, \theta)}{\partial \beta} = \frac{n}{\beta} + n \ln \theta - \sum_{i=1}^{n} \ln x_i = 0 \quad \beta^* = \frac{n}{\sum \ln x_i - n \ln \theta}$$

$$\frac{\partial^2 \ln L(\beta, \theta)}{\partial^2 \beta} = -\frac{n}{\beta^2} < 0 \Rightarrow \beta^* = \arg\max$$

б) β - известно

$$\frac{\partial \ln L(\beta, \theta)}{\partial \theta} = \frac{n\beta}{\theta} > 0$$, т.е чем больше θ, тем более правдоподобно. Но есть ограничение: $\theta < X_i \Rightarrow$

$\Rightarrow \theta^* = X_{(1)}$

в) Оба параметра неизвестны

$\theta^* = X_{(1)}$

$$\beta^*\Big|_{\theta^*} = \frac{n}{\sum \ln x_i - n \ln X_{(1)}}$$

№11

Пусть у нас есть независимая последовательность испытаний Бернулли. Обозначим X – количество успехов, n – количество всех испытаний.

$$\gamma^* = \frac{X}{n}$$

Проверим несмещенность:

$$E\gamma^* = E\left(\frac{X}{n}\right) = \frac{EX}{n} = \frac{np}{n} = p \text{ – несмещ.}$$

$$EX = \sum_{k=1}^{n} P(X=k)\cdot k = \sum_{k=1}^{n} k\cdot C_n^k p^k (1-p)^{n-k} =$$

$$f(x) = \sum_{k=1}^{n} C_n^k x^k q^{n-k} = (x+q)^n$$

$$x\cdot f'(x) = \sum_{k=1}^{n} k C_n^k x^k q^{n-k}$$

$$EX = x\cdot f'(x)\Big|_{x=p} = x\cdot n(x+q)^{n-1}\Big|_{x=p} = p\cdot n\cdot(p+1-p)^{n-1} = np \Rightarrow$$

$$\Rightarrow EX = np$$

Эффективность:

$$\cancel{I(p) = E\left(\frac{\partial}{\partial p}\ln L(p)\right)^2}$$

$$\cancel{L(p) = P(X_1=x_1)\ldots P(X_n=x_n) = p^{\sum x_i}(1-p)^{n-\sum x_i}}$$

$$I(p) = E\left(\frac{\partial}{\partial p}\ln P(p,X)\right)^2 = E\left(\frac{\partial}{\partial p}\ln C_n^X p^X (1-p)^{n-X}\right)^2 =$$

$$= E\left(\frac{\partial}{\partial p}\left(\ln C_n^X + X\ln p + (n-X)\ln(1-p)\right)\right)^2 = E\left(\frac{X}{p} - \frac{n-X}{1-p}\right)^2 =$$

$$= E\left(\frac{X^2}{p^2} + \frac{(n-X)^2}{(1-p)^2} - \frac{2X(n-X)}{p(1-p)}\right) = \frac{1}{p^2(1-p)^2}E\left(X^2(1-p)^2 + (n-X)^2\cdot\right.$$

$$\left.\cdot p^2 - 2X(n-X)(1-p)p\right) = \frac{1}{p^2(1-p)}E\left(X^2 - 2pX^2 + p^2X^2 + n^2p^2 -\right.$$

$$\left.- 2nXp^2 + X^2p^2 - 2Xnp + 2X^2p + p^2\cdot 2Xn - 2X^2p^2\right) = \frac{1}{p^2(1-p)^2}\cdot$$

$$\cdot E\left(X^2 - 2Xnp + n^2p^2\right) = \frac{1}{p^2(1-p)}E(X-np)^2 = \frac{1}{p^2(1-p)^2}\cdot DX =$$

$$= \frac{np(1-p)}{p^2(1-p)^2} = \frac{n}{p(1-p)}$$

$$\frac{1}{I(p)} = \frac{p(1-p)}{n} \quad \left(\frac{1}{I(p)\cdot n} \text{ – для выборки длины } n\right)$$

$$E(\gamma^* - \theta)^2 = E\left(\frac{X}{n} - p\right)^2 = \cancel{D\left(\frac{X}{n}\right) = \frac{1}{n^2}DX = \frac{np(1-p)}{n^2}} =$$

$$= \cancel{\frac{p(1-p)}{n^2}} = D\left(\frac{X}{n}\right) = \frac{1}{n^2}DX = \frac{np(1-p)}{n^2} = \frac{p(1-p)}{n} =$$

$$= \frac{1}{I(p)} \Rightarrow \text{оценка является эффективной}$$

№6

$\vec{X} = (X_1, \ldots, X_n)$ из $N(a, \sigma^2)$

$T(x) = \frac{\alpha}{n}\sum_{k=1}^{n}|X_k - a|$ - несмещен и сильно состоятельная оценка параметра σ.

$\alpha = ?$

1) $T(x)$ несмещен $\Leftrightarrow$ $ET(x) = \sigma$

2) $T(x)$ сильно состоят. $\Leftrightarrow$ $T(x) \xrightarrow{P} \sigma$

1) $ET(x) = E\left(\frac{\alpha}{n}\sum_{k=1}^{n}|X_k - a|\right) = \frac{\alpha}{n}\sum_{k=1}^{n}E|X_k - a|$

$f_{X_k} = \frac{1}{\sqrt{2\pi\sigma^2}}\exp\left(-\frac{(x-a)^2}{2\sigma^2}\right)$, $f_{X_k - a} = \frac{1}{\sqrt{2\pi\sigma^2}}\exp\left(-\frac{x^2}{2\sigma^2}\right)$

$F_{|X_k - a|}(x) = P(|X_k - a| < x) = P(X_k - a < x, X_k - a > -x) =$

$= \int_{-x}^{x} f_{X_k - a}(x)\,dx = \int_{-x}^{x}\frac{1}{\sqrt{2\pi\sigma^2}}\exp\left(-\frac{x^2}{2\sigma^2}\right)dx = 2\int_{0}^{x}\frac{1}{\sqrt{2\pi\sigma^2}}e^{-\frac{x^2}{2\sigma^2}}dx$

$f_{|X_k - a|}(x) = \frac{d}{dx}F_{|X_k - a|}(x) = 2\cdot\frac{1}{\sqrt{2\pi\sigma^2}}\exp\left(-\frac{x^2}{2\sigma^2}\right)I\{x \geq 0\}$

$ET(x) = E|X_k - a| = 2\int_{0}^{\infty}\frac{1}{\sqrt{2\pi\sigma^2}}e^{-\frac{x^2}{2\sigma^2}}\cdot x\,dx =$

$I = 2\int_{0}^{\infty}\frac{1}{\sqrt{2\pi\sigma^2}}e^{-\frac{x^2}{2\sigma^2}}x\,dx = \sqrt{\frac{2}{\pi\sigma^2}}\int_{0}^{\infty}d\,e^{-\frac{x^2}{2\sigma^2}}\cdot\frac{2\sigma^2}{-2} = -\sqrt{\frac{2\sigma^2}{\pi}}\int_{0}^{\infty}d\,e^{-\frac{x^2}{2\sigma^2}} =$

$= -\sqrt{\frac{2}{\pi}}\sigma\cdot e^{-\frac{x^2}{2\sigma^2}}\Big|_{0}^{\infty} = \sqrt{\frac{2}{\pi}}\sigma$

$E(T(x)) = \frac{\alpha}{n}\sum_{k=1}^{n}\sqrt{\frac{2}{\pi}}\sigma = \alpha\cdot\sqrt{\frac{2}{\pi}}\sigma = \sigma \Rightarrow \alpha = \sqrt{\frac{\pi}{2}}$

2) $T(x) \xrightarrow{P} \sigma$; $\frac{\alpha}{n}\sum_{k=1}^{n}|X_k - a| \longrightarrow \alpha\cdot\frac{|X_1 - a| + |X_2 - a| + \ldots + |X_n - a|}{n} \longrightarrow$

$\xrightarrow{\text{закон больших чисел в форме Хинчина}} \alpha\cdot E|X_k - a| = \alpha\cdot\sqrt{\frac{2}{\pi}}\sigma = \sqrt{\frac{\pi}{2}}\cdot\sqrt{\frac{2}{\pi}}\sigma = \sigma$ - состоят.

Ответ: $\alpha = \sqrt{\frac{\pi}{2}}$

№7

$P(X = k) = \frac{1}{2}\left(\frac{\lambda_1^k}{k!}e^{-\lambda_1} + \frac{\lambda_2^k}{k!}e^{-\lambda_2}\right)$

$0 < \lambda_1 < \lambda_2$

a) $EX = \sum_{k=1}^{\infty}k\cdot P(X = k) = \frac{1}{2}\sum_{k=1}^{\infty}k\left(\frac{\lambda_1^k}{k!}e^{-\lambda_1} + \frac{\lambda_2^k}{k!}e^{-\lambda_2}\right) =$

$= \frac{1}{2}\left(\sum_{k=1}^{\infty}k\cdot\frac{\lambda_1^k}{k!}e^{-\lambda_1} + \sum_{k=1}^{\infty}k\cdot\frac{\lambda_2^k}{k!}e^{-\lambda_2}\right) = \frac{1}{2}(\lambda_1 + \lambda_2)$

№10

$\overline{X}$ - эффективная оценка матожидания в классе линейн. несмещ. функций - док-ть

$E\overline{X} = EX_i$ - сама эффективна

$a^* = \sum_{i=1}^{n} \alpha_i X_i$

$Ea^* = E\left(\sum_{i=1}^{n} \alpha_i X_i\right) = \sum_{i=1}^{n} \alpha_i E(X_i) = \sum_{i=1}^{n} \alpha_i E(X_1) =$

$= E(X_1) \cdot \sum \alpha_i =$

$Ea^* = EX_1$, $\sum_{i=1}^{n} \alpha_i = 1$ - класс лин. несмещ. оценок должен удовлетворять этому условию

$EX_1 = a$

$E(a^*-a)^2 \to \min$; $E(a^*-a)^2 = E\left(\sum_{i=1}^{n} \alpha_i X_i - a\right)^2 = E\left(\left(\sum_{i=1}^{n}\alpha_i X_i\right)^2 + a^2 - 2a\sum_{i=1}^{n}\alpha_i X_i\right) = a^2 - 2a\sum_{i=1}^{n}\alpha_i EX_i + E\left(\sum_{i=1}^{n}\alpha_i X_i\right)^2 = a^2 - 2a^2 + E\left(\sum_{i=1}^{n}\alpha_i X_i\right)^2 = E\left(\sum_{i=1}^{n}\alpha_i X_i\right)^2 - a^2 = E\left(\sum_{i=1}^{n}\alpha_i^2 X_i^2 + \sum_{i\neq j}\alpha_i\alpha_j X_i X_j\right) - a^2 =$

$= \sum_{i=1}^{n}\alpha_i^2 EX_i^2 + \sum_{i\neq j}\alpha_i\alpha_j EX_i EX_j - a^2 = a^2\sum_{i\neq j}\alpha_i\alpha_j - a^2 + \sum_{i=1}^{n}\alpha_i^2(EX_i^2 - a^2 + a^2) =$

$= a^2\sum_{i\neq j}\alpha_i\alpha_j - a^2 + \sum_{i=1}^{n}\alpha_i^2 DX_i + a^2\sum_{i=1}^{n}\alpha_i^2 = a^2\left(\sum\alpha_i\right)^2 - a^2 + \sum_{i=1}^{n}\alpha_i^2 DX_i =$

$= \sum_{i=1}^{n}\alpha_i^2 DX_i = \sum_{i=1}^{n}\alpha_i^2 DX_1 = DX_1 \cdot \sum_{i=1}^{n}\alpha_i^2$

$\begin{cases} \sum\alpha_i = 1 \\ \sum\alpha_i^2 \to \min \end{cases}$ $\quad \sum_{i=1}^{n}\alpha_i = \sum_{i=1}^{n-1}\alpha_i + \alpha_n = 1$, $\alpha_n = 1 - \sum_{i=1}^{n-1}\alpha_i$

$\sum_{i=1}^{n}\alpha_i^2 = \sum_{i=1}^{n-1}\alpha_i^2 + \left(1 - \sum_{i=1}^{n-1}\alpha_i\right)^2 = f(\alpha_1, \alpha_2, \dots \alpha_{n-1})$

Минимум: необх. усл: $\forall i \ \frac{\partial f}{\partial \alpha_i} = 0$;

α_k: $\frac{\partial f}{\partial \alpha_k} = 2\alpha_k + 2\left(1 - \sum_{i=1}^{n-1}\alpha_i\right)\cdot(-1) = 0 \quad \alpha_k = 1 - \sum_{i=1}^{n-1}\alpha_i$

α_j: $\frac{\partial f}{\partial \alpha_j} = 2\alpha_j + 2\left(1 - \sum_{i=1}^{n-1}\alpha_i\right)(-1)$, $\alpha_j = 1 - \sum_{i=1}^{n-1}\alpha_i \Rightarrow \alpha_k = \alpha_j$

k и j - произвольные $\alpha_1 = \alpha_2 = \dots = \alpha_{n-1} = \alpha_n$

$\frac{\partial^2 f}{\partial \alpha_k^2} = 2 + 2 = 4$, $\frac{\partial^2 f}{\partial \alpha_k \partial \alpha_j} = 2$, все главные миноры $> 0 \Rightarrow$

$\Rightarrow$ действит. минимум

$\alpha_1 = \alpha_2 = \dots = \alpha_{n-1} = \alpha_n \Rightarrow \alpha_i = \frac{1}{n} \Rightarrow$ эффективная оценка

$a^*_{эф} = \sum_{i=1}^{n}\alpha_i X_i = \sum_{i=1}^{n}\frac{1}{n}X_i = \frac{1}{n}\sum_{i=1}^{n}X_i = \overline{X}$ ч.т.д.